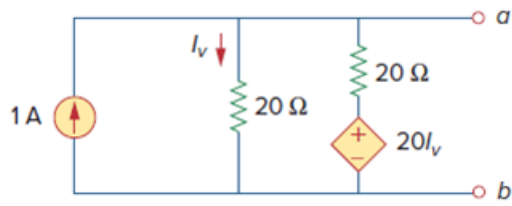


Problem

Obtain the Thevenin and Norton equivalent circuits of the circuit in Fig. 4.114 with respect to terminals a and b .

Figure. 4.114:



Step-by-step solution

Step 1 of 7

Refer to circuit diagram in Figure 4.114 in the text book.

Consider the circuit diagram as show in Figure 1 to calculate the Thevenin's voltage.

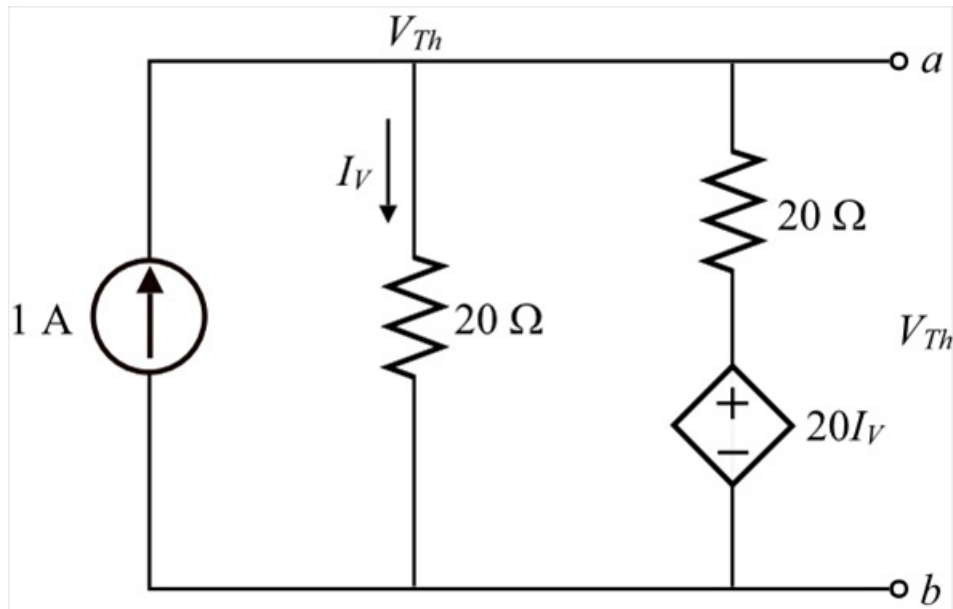


Figure 1

Step 2 of 7

The current I_v is,

$$I_v = \frac{V_{Th}}{20}$$

Apply Kirchhoff's Current law at node 1.

$$\frac{V_{Th} - 20I_v}{20} + \frac{V_{Th}}{20} = 1$$

Substitute $\frac{V_{Th}}{20}$ for I_v .

$$\frac{V_{Th} - 20\left(\frac{V_{Th}}{20}\right)}{20} + \frac{V_{Th}}{20} = 1$$

$$\left(\frac{1-1}{20} + \frac{1}{20}\right)V_{Th} = 1$$

$$\left(\frac{1}{20}\right)V_{Th} = 1$$

$$V_{Th} = 20 \text{ V}$$

Therefore, the value of the Thevenin's voltage V_{Th} is **20 V**.

Step 3 of 7

To calculate the Norton's current, short circuit the load terminals. The modified circuit is shown in Figure 2.

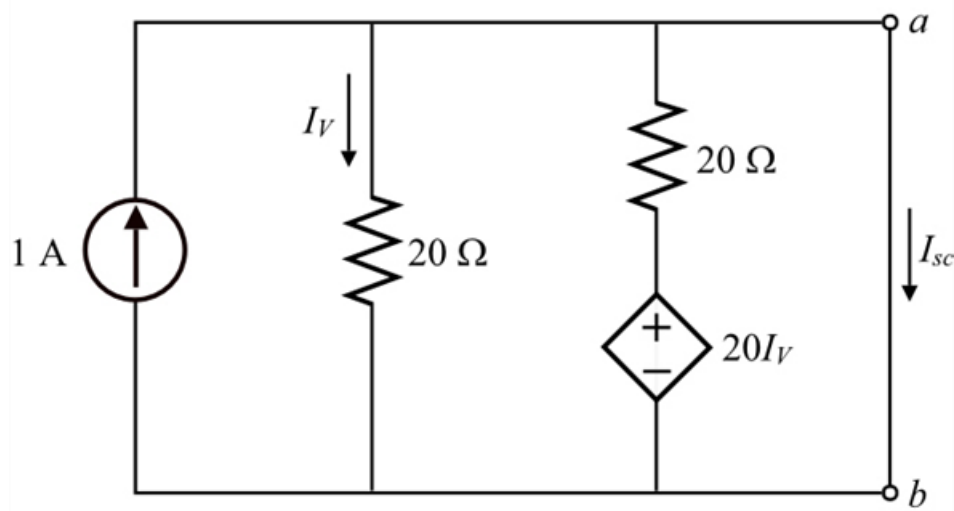


Figure 2

Step 4 of 7

The value of the short circuit current is equal to the source current since the output terminals are in parallel to the source. No current flows through the resistances in the circuit.

Hence, the value of the short circuit current I_{sc} is **1 A**.

Step 5 of 7

Calculate the value of the value of the Thevenin's resistance or Norton's resistance.

$$R_{Th} = \frac{V_{Th}}{I_{sc}}$$

Substitute 1 A for I_{sc} and V_{Th} for 20 V .

$$\begin{aligned} R_{Th} &= \frac{20}{1} \\ &= 20\ \Omega \end{aligned}$$

Therefore, the value of the Thevenin's resistance or Norton's resistance is $20\ \Omega$.

Step 6 of 7

Draw the Thevenin's equivalent circuit diagram.

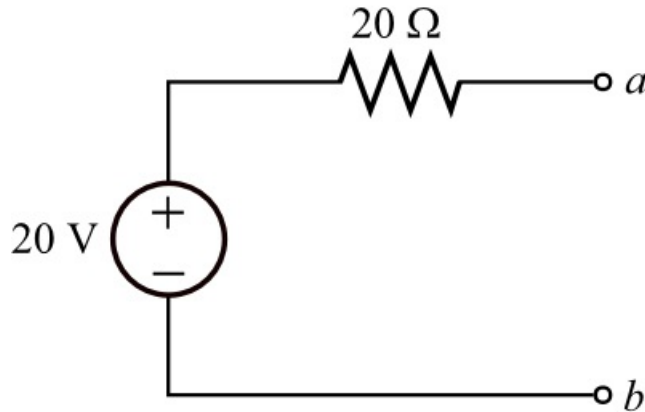


Figure 3: Thevenin's circuit

Therefore, the Thevenin's equivalent circuit diagram is shown in Figure 3.

Step 7 of 7

Draw the Norton's equivalent circuit diagram.

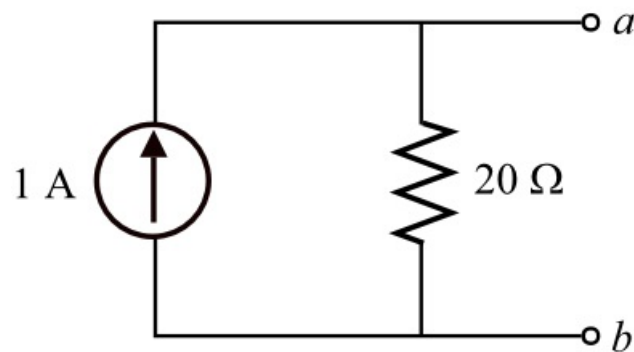


Figure 4: Norton's circuit

Therefore, the Norton's equivalent circuit diagram is shown in Figure 4.